

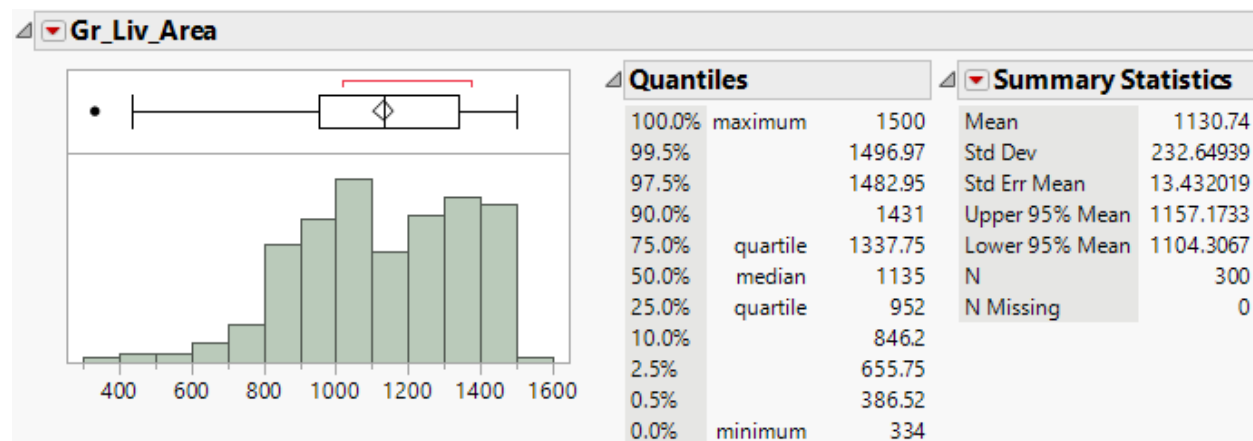
Predictive Model for Housing Prices Using Simple Linear Regression for the Story County Assessor's Office in Ames, Iowa

Introduction

In this report, we will be fulfilling the request of our client, Story County Assessor's Office in Ames Iowa, which is responsible for determining property value for tax purposes. We have been hired to assist in developing a predictive model for estimating housing prices based on key property characteristics, and for this project, we will be focusing on they key property characteristic of above-ground living area, notated as "Gr_Liv_Area" in our dataset. Our data set contains 300 data points, with each data point representing one property. We will be using simple linear regression to predict housing prices, notated as "SalePrice" in our dataset, based on the above-ground living area. We will be assessing how well the predictor, above-ground living area, explains variations in housing prices, as well as providing recommendations to the Story County Assessor's Office on how this model can be used to predict and assess property values. We will be using the JMP software in order to analyze and visualize our data.

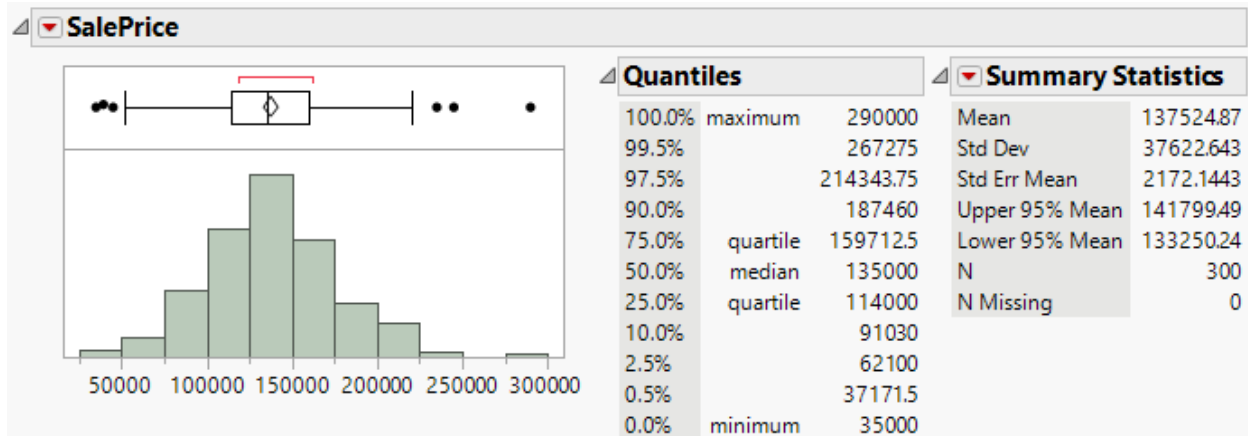
Data Understanding

Figure 1 – Key Statistics of Above-Ground Living Area



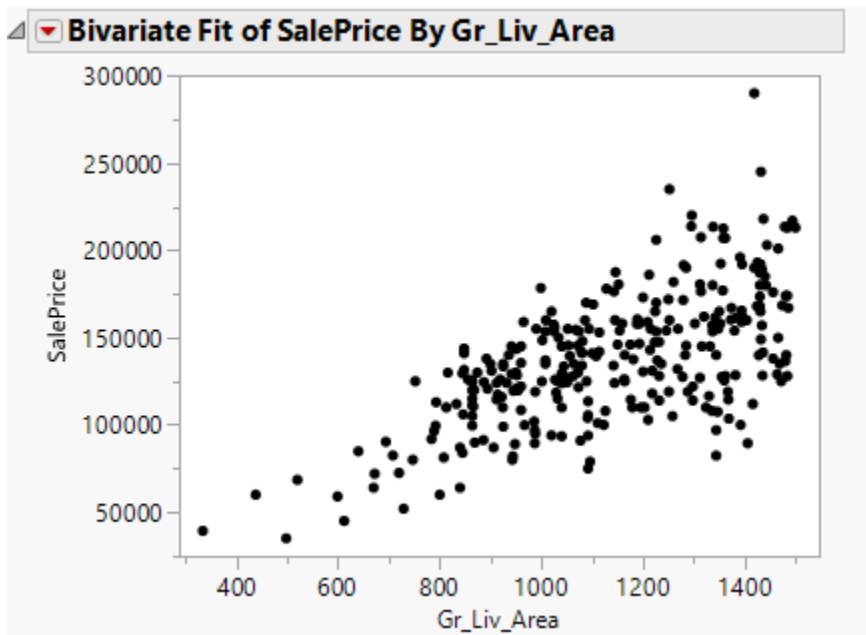
In this figure, we see the key statistics of the variable "Gr_Liv_Area", which represents the above-ground living area. In our dataset, this variable holds a mean of 1,130.74, a median of 1,135, a standard deviation of 232.64939, an interquartile range that is calculated to be 385.75, as well as a minimum-to-maximum range that ranges from 334 to 1,500. The histogram and boxplot both help to visualize the distribution of the above-ground living area values.

Figure 2 – Key Statistics of Housing Prices



In this figure, we see the key statistics of the variable “SalePrice”, which represents the housing prices. In our dataset, this variable holds a mean of 137,524.87, a median of 135,000, a standard deviation of 37,622.643, an interquartile range that is calculated to be 45,712.5, as well as a minimum-to-maximum range that ranges from 35,000 to 290,000. The histogram and boxplot both help to visualize the distribution of the housing prices.

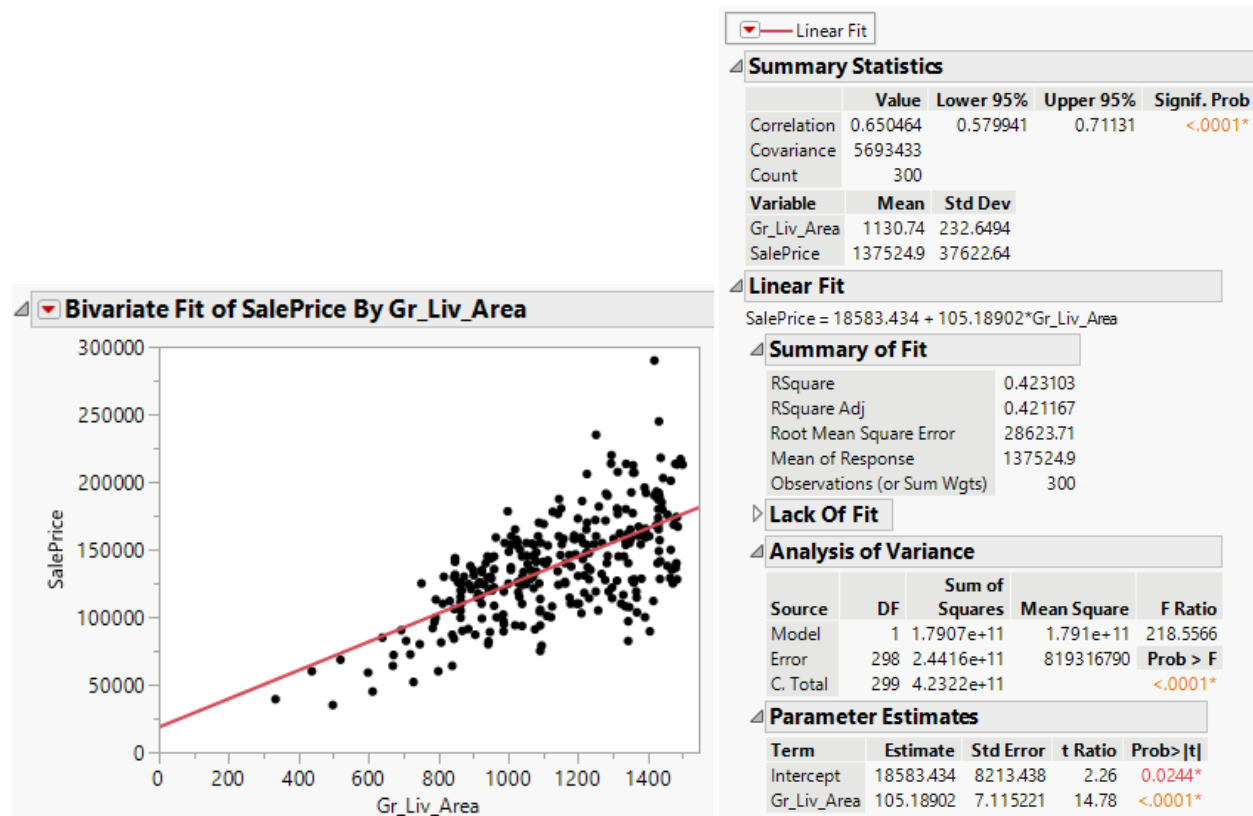
Figure 3 – Scatter Plot of Above-Ground Living Area vs. Housing Price



In this figure, we see a scatter plot that shows the relationship between the above-ground living area and the housing price. At an initial glance, there seems to be somewhat of a positive correlation between the two variables, but we will not make any definitive conclusions until statistical analysis has been done to assess this relationship.

Simple Linear Regression Analysis

Figure 4 – Simple Linear Regression Model – Using Above-Ground Living Area to Predict Housing Price



In this figure, we see a scatter plot displaying the relationship between above-ground living area and housing price, along with a visualization of the linear relationship between the two variables.

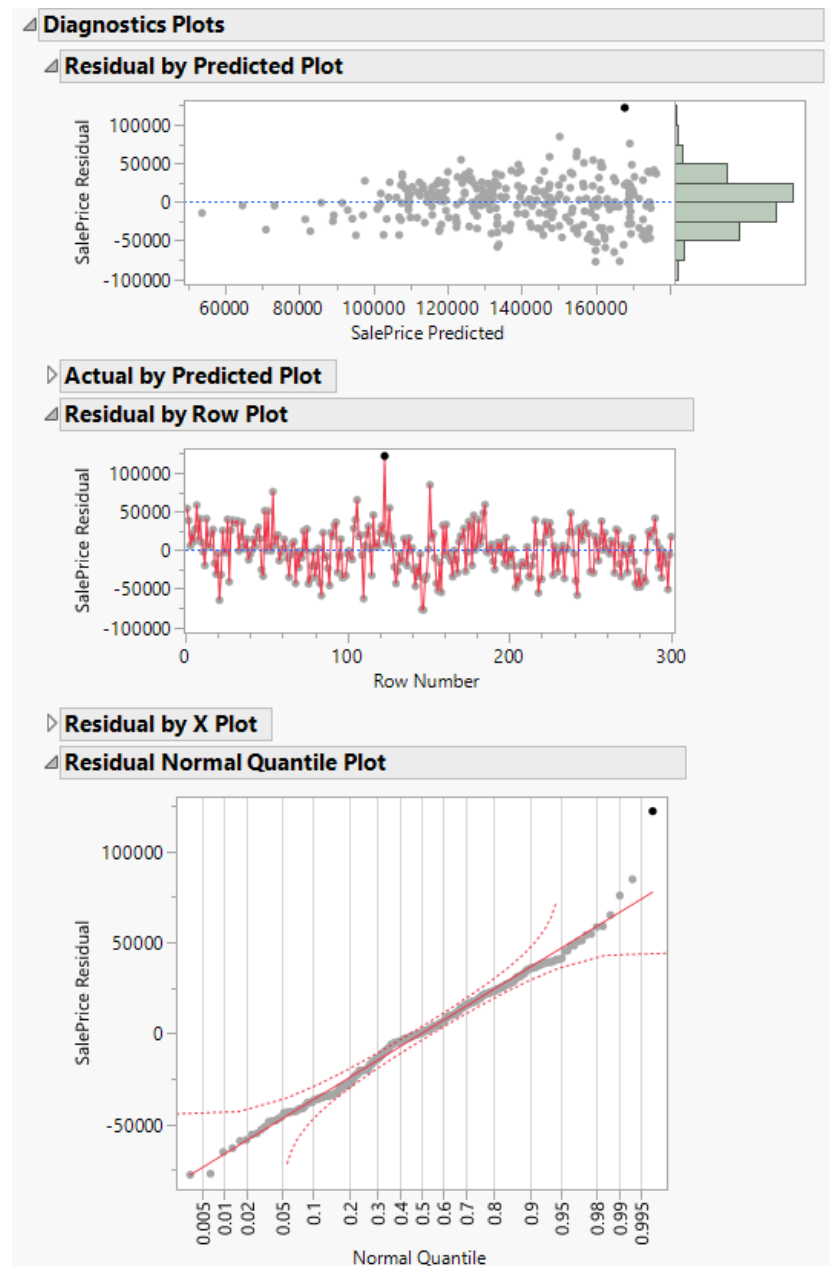
The regression equation is given below the “Linear Fit” headline as

$\text{SalePrice} = 18583.434 + 105.18902 \times \text{Gr_Liv_Area}$. The slope is therefore 105.18902, and the intercept is 18,583.434. The slope indicates that for every one unit increase of Gr_Liv_Area, there is a 105.18902 unit increase of SalePrice. The intercept indicates that when the value of Gr_Liv_Area is zero, the SalePrice will have a value of 18,583.434. The R-squared value of 0.423103 measures the proportion of the total variation in SalePrice that is explained by Gr_Liv_Area and this regression model, and indicates that **about 42% of the variation in SalePrice can be explained by Gr_Liv_Area using the regression model**. The R-squared value indicates a moderate level of explanation of the variability in SalePrice using this regression model. The correlation coefficient is given as 0.650464, which indicates **a moderately strong, positive linear relationship between the two variables**. The F-statistic, denoted as F Ratio, is at a high value of 218.5566, suggesting that there would likely be a

significant difference in effectiveness between the regression model and a baseline model. The p-value is less than 0.0001, which is clearly below a standard alpha value of 0.05. Thus, we would reject the null hypothesis that the regression model is as effective at providing a fit as a baseline model that simply uses the average SalePrice, and conclude that the regression model does provide a better fit than such a baseline model.

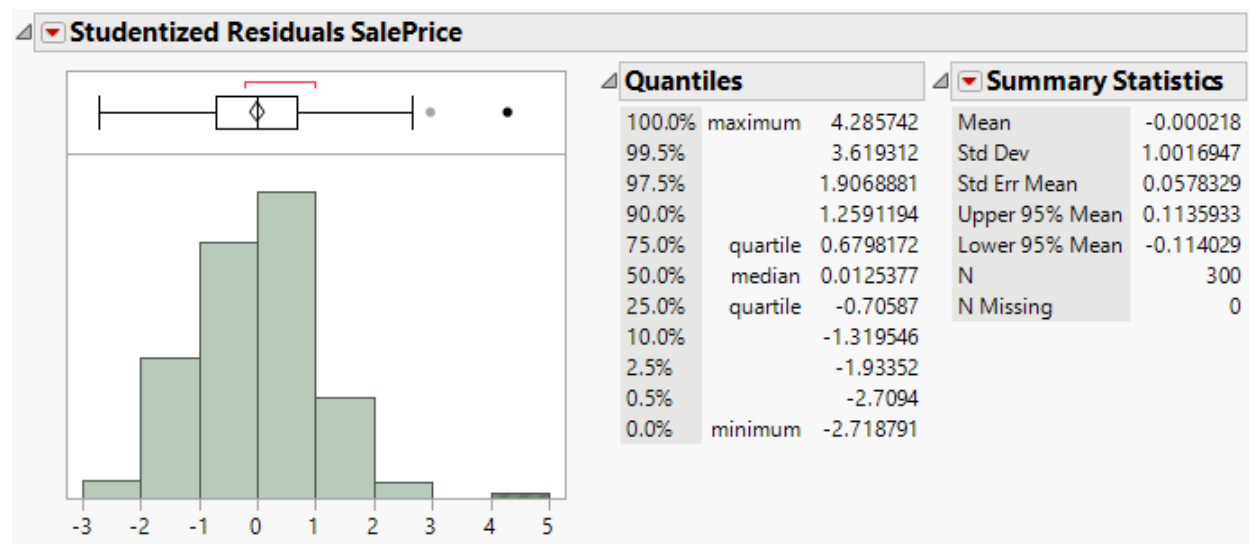
Assumptions and Model Diagnostics

Figure 5 – Residual Plots



In this figure, we see three residual plots: the Residual by Predicted Plot, which shows the residuals plotted against the predicted values of Sale Price, the Residual by Row Plot, which shows the residuals plotted against row numbers, and the Residual Normal Quantile Plot, which shows the quantiles of the residuals plotted against the quantiles of a standard normal distribution. Both the Residual by Predicted Plot and the Residual by Row Plot display a random scattering of data points around the blue-dotted line where the residual equals zero, indicating that the model is generally effective at capturing the relationship between the variables of interest. This random scattering of the residuals around the line where the residual equals zero also indicates that the residuals will have a roughly constant variance, fulfilling the requirement of **homoscedasticity**, be roughly linear, fulfilling the requirement of **linearity**, be roughly **normally distributed**, as shown by the histogram accompanying the Residual by Predicted Plot, and be largely **independent of one another**. However, there is a single observation in both, marked by a dark black dot, that seems to deviate relatively far from the line where the residual equals zero. The Residual Normal Quantile Plot displays a roughly linear plot, indicating that the residuals are roughly normally distributed, but also displays patterns that curve from the line, indicating that the residuals are not fully normally distributed. The Residual Normal Quantile Plot also shows a single observation, marked a dark black dot, that seems to deviate from the rest of the residuals. We will look further into this outlier in Figure 6.

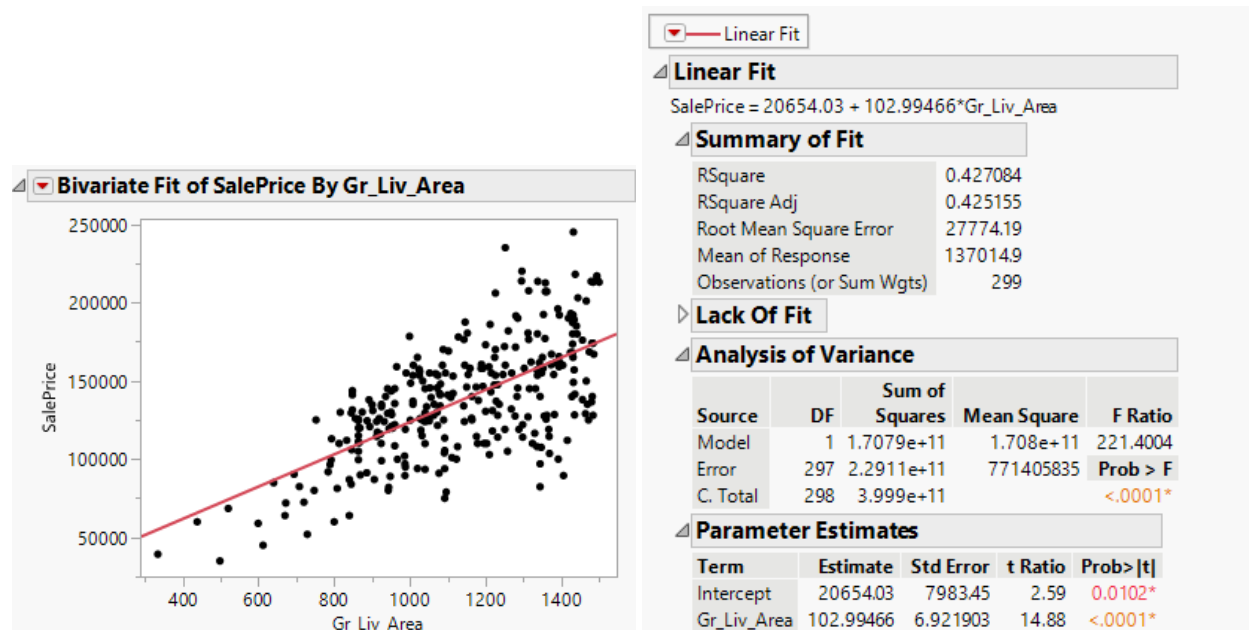
Figure 6 – Studentized Residuals



In this figure, we see the distribution of the studentized residuals. One observation is distinguished above the others: the residual maximum value of 4.285742, marked by a dark

black dot in the box plot. This is the only observation that has a residual value beyond ± 3 standard deviations of the mean, and thus may likely be an outlier.

Figure 7 – Simple Linear Regression Model Without Outlier



In this figure, we re-run the regression model after removing the outlier that was identified in Figures 5 and 6. When comparing this figure with the original regression model in Figure 4, we observe that the slope and intercept of the regression equation have not significantly changed, from 18,583.434 to 20,654.03 and from 105.18902 to 102.99466 respectively, and that the R-squared value has also scarcely changed, from 0.423103 to 0.427084. The p-value remains at <0.0001 for this new regression model. This indicates that removing the outlier did not change the model significantly, and thus that there should not be any violations that significantly affect the accuracy of our predictive model. We can therefore proceed with applying the regression model to our study of the relationship between above-ground living area and housing price.

Interpretations and Recommendations (Managerial Summary)

In conclusion, we believe that this regression model is significantly effective in predicting housing prices based on above-ground living area. We saw in Figure 4 that there was a R-squared value of 0.423103, indicating that roughly 42% of the variation in housing prices could

be explained by the above-ground living area using the regression model. Although this is not an extremely high percentage, it still shows that the above-ground living area can be used by the regression model to provide a moderate level of explanation for the variability in the housing prices. The correlation coefficient of 0.650464 from Figure 4 indicates a moderately strong, positive linear relationship between the two variables. These statistics give credibility to the regression equation of $\text{SalePrice} = 18583.434 + 105.18902 * \text{Gr_Liv_Area}$ in Figure 4. The slope indicates that for every one-unit increase in above-ground living area, the housing price increases by roughly \$105.19. We understand from the positive slope that in general, an increased above-ground living area should correspond to an increased housing price. The residual plots in Figure 5 show that the residuals are roughly randomly scattered, and thus that the key assumptions of linearity, homoscedasticity, normality, and independence of residuals should be generally fulfilled. The diagnostics testing shown in Figures 6 and 7 where an outlier was identified and excluded from the regression model shows that the outlier does not have a significant effect on the regression model, and therefore that the regression model can be used without additional steps to remove outliers.

In terms of the regression model's utility, the regression equation can be used to make ballpark estimates on housing prices using the property's above-ground living area. The statistics seem to indicate that the model is at least moderately effective at capturing a relationship between the two variables, and therefore the regression equation should have a degree of value when predicting housing prices using the above-ground living area. However, a few limitations exist with the model as well. The intercept of the regression equation is 18,583.434, representing the housing price when the above-ground living area is zero. Realistically, a property can not have an above-ground living area of zero. If the model were to be applied to unique properties with a relatively very small above-ground living area, such as underground properties, it may lose its accuracy. Additionally, although the R-squared value of roughly 0.42 is not negligible, it is not very high either. It should be noted that there is still a large proportion of variability in housing price that is not accounted for by simply using the above-ground living area with this model, and therefore the recommendation is that additional predictor variables, such as the property's vicinity to a public school, be incorporated into the regression model to predict housing prices.

Overall, this regression model should be moderately effective at predicting housing prices based on above-ground living area. It could potentially even be highly effective if additional predictor variables were incorporated into the model, in order to account for even more of the variability in housing prices.